

Perceptual uncertainty, the distance–time anomaly, and the missing lateral term

*2026-08-28. Written after two findings that the current field cannot explain — the LTAP speed effect at matched PET, and the cyclist overtake's lateral criticality — and after reading five papers Jonas pointed toward or that the search turned up. The claim of this note is that both follow from one structural property of the field — that it evaluates the deficit along a single predicted trajectory — and that the natural repair is native to active inference.

[Corrected 2026-08-28, later the same day, while explaining §5 to Jonas. An earlier version of this line claimed the two findings are "one problem, not two" with a single fix. Working the sign through, that is too strong. The lateral case is safe: the deficit is flat at wide clearance and rises steeply at small clearance, so averaging over a spread raises it, and raises it more when the mean clearance is small — the right direction. The distance case is not safe: if predictive variance grows with distance and the deficit is convex, then the more distant conflict gets the higher expected deficit, i.e. more discomfort at 70 km/h — the opposite of what we observe. So the expectation route earns the lateral term but does not, on its own, explain the LTAP effect; that likely needs the bounded-optimal route of Wang et al., where noise changes the decision rule rather than only the evaluation. §5's proposal stands for the lateral problem and is downgraded to a candidate for the distance problem. This is exactly the derivation §6 puts first, and it is why it goes first.] Companion documents: docs/overtake_construction_note.md (where both findings came from), docs/lane_entry_note.md (the lateral machinery as it stands), docs/active_inference_for_czb_assessment.md (the framework argument).*

1 The two things the field cannot currently do

The LTAP speed effect. In the Random LTAP design, intervention is *lower* at 70 km/h than at 50 km/h at every one of the nine PET levels (at PET2: 0.360 against 0.605).

The manipulation is worth stating precisely, because it decides what the effect can mean. Checked in the traces: the vehicle whose speed changes between the two conditions is the **oncoming** one (2.01 m wide, 13.9 m/s in the 50 kph clips and 19.4 m/s in the 70 kph clips); the other vehicle holds 13.9 m/s in both. So the ego's own kinematics are identical across the manipulation, and PET — the arrival-time separation at the conflict zone — is matched by design.

That makes the effect unusually clean. If the time-based conflict measure is matched and the ego's own state is unchanged, then essentially the only thing that differs between the conditions is **how far away the oncoming vehicle is** at any moment, since a faster vehicle covers the same time gap over a longer distance. The whole effect is therefore carried by distance, at matched time. Our field is built from time-like quantities — inverse tau, TTC, required deceleration — which are by construction close to identical here, so it predicts almost no effect, and a large one is present.

This is also exactly the manipulation used in the gap-acceptance literature below, where the oncoming vehicle's speed is varied at matched time-to-arrival; our result is a replication of a documented phenomenon rather than an anomaly of this dataset.

The cyclist-overtake lateral effect. Passing a cyclist at 0.5 m rather than 1.5 m roughly quadruples the intervention rate, while in *all* conditions the pass is collision-free. The field's lateral machinery only asks whether a collision geometry applies; it answers "yes, fully" throughout, and so cannot see the manipulation (overtake_construction_note.md §4).

These look unrelated. Section 5 argues they are the same missing ingredient.

2 What the literature says about the distance–time anomaly

Zgonnikov, Abbink and Markkula (2024) modelled left-turn gap acceptance — our LTAP scenario — with a generalized drift-diffusion model, and found that the probability of going increases with **both** time-to-arrival **and** distance gap. Their model needs a *generalized gap measure* combining the two; pure time-to-arrival does not suffice. Response time increased with time-to-arrival, and there was an additional effect of distance on "wait" responses. So the phenomenon we observe is documented, in our scenario, by this group.

Wang, Srinivasan, Jokinen, Oulasvirta and Markkula (2024) supply the mechanism Jonas remembered, in the pedestrian-crossing version of the same problem. Their model is bounded-optimal decision-making under *noisy visual perception*: the agent estimates distance and speed through a visual system whose angular noise translates into

distance-dependent uncertainty, so estimates of time-to-arrival become more dispersed in some conditions than others. Because the cost of a misjudgement is asymmetric, the rational response to a noisier estimate is a more conservative decision. Their model reproduces four phenomena, of which the second is exactly ours: **greater gap acceptance at higher speed for the same time-to-arrival**. Their framing is worth quoting for the project's own argument: on this account speed-dependent gap acceptance "is an entirely rational behaviour" given the visual system's constraints, not a bias to be corrected.

Mohammad, Farah and Zgonnikov (overtaking gap acceptance) tried the same family on a third scenario and concluded that the model needs an **initial decision bias that depends on the initial velocity** — a speed-dependent starting point, rather than a speed-dependent drift or boundary. That is a third independent report that a speed or distance term has to enter *somewhere* beyond the time-to-collision variable, and it is a useful warning that where it enters is not obvious: the same qualitative effect can be produced by a starting bias, a drift term, or a boundary.

Bontje, van Waveren, van Maanen, Nallapu, Markkula and Zgonnikov (2026) is a semi-systematic review of 28 evidence-accumulation studies in traffic (2014–2026). Two things in it bear on us directly. It confirms that in driving applications the drift is conventionally driven by **looming (optical expansion) or TTC** — not, as in our card A.3 accumulator, by a comfort-zone deficit; and it lists **leaky accumulation** and **urgency/collapsing boundaries** as the standard architectures for exactly the failure modes we hit. Our A.3 accumulator used none of these. That does not overturn the FAIL — which was pre-registered and stands — but it does say the FAIL was of one specific accumulator, not of the accumulator family, and the reserve remedy we held back (leaky accumulation) is the field's standard first move.

3 What the literature offers for the lateral problem

Kolekar, de Winter and Abbink (2020) propose the **Driver's Risk Field (DRF)**, and it is close enough to our construction to be worth taking seriously as a donor rather than a rival. The DRF is a two-dimensional field around the car representing the driver's belief about the probability of an event at each point; multiplied by the *consequence* assigned to what is at that point and summed, it gives a scalar perceived risk. Human-like behaviour then emerges from a controller that **keeps that scalar below a threshold**.

Two features matter for us. First, the shape: the field is a torus with a Gaussian cross-section along the predicted path, whose height grows as a parabola out to a look-ahead distance that scales with speed, and whose **width grows with arc length and with the absolute steering angle** — the latter justified as signal-dependent noise in the sensorimotor system. Width at the vehicle is set to car-width/4, so $\pm 2\sigma$ spans the car. Second, the standing: it is parameterized by six constants ($p, t_{la}, m, c, k1, k2$) that depend only on the driver's state and not on the environment, and it was shown to reproduce behaviour across seven scenarios including obstacle avoidance, car-following and **overtaking** — the two scenarios we are currently failing to unify.

The structural point: "keep a scalar risk below a threshold" is a comfort-zone boundary on a scalar, which is our claim exactly. The DRF is, in effect, an independently validated instance of the framing this project is testing — with the lateral dimension already built in, and validated on overtaking.

4 Why our field lacks both, structurally

Our preference field evaluates the deficit **along a single predicted trajectory**: the ego's actual motion, with the other vehicle's motion projected forward. Everything is a point estimate. The lane-entry weight is the one place where a distribution appears, and even there the "overlap fraction" is a geometric overlap of two rigid width intervals, not a probability.

A point-estimate field has two unavoidable consequences, and they are precisely our two failures. It cannot express that a *clear* pass is uncomfortable, because on the predicted trajectory nothing happens — you need probability mass off the predicted path for a 0.5 m clearance to cost more than a 1.5 m one. And it cannot express that a distant vehicle is less certain than a near one, because a point estimate has no width — you need the estimate's dispersion to grow with distance for the speed effect at matched PET to appear.

5 One proposal that addresses both, and is native to active inference

Give the predicted trajectory a distribution, and take the deficit in expectation over it. Concretely: replace the deficit evaluated at the predicted relative state, $d(x_{\blacksquare})$, with $E[d(x)]$ where $x \sim N(x_{\blacksquare}, \Sigma(t))$ and Σ grows with prediction horizon and with distance.

This is not a bolt-on. Expected free energy is *already* an expectation of a preference term under a predictive distribution — the released model simply collapses that distribution to its mean for the scenarios it was built for, which are longitudinal and where the collapse is harmless. Restoring the width is a return to the framework's own form, and it is the same move as our lane-entry work (§3 of the lane-entry note argued the continuous form is the closed-form of an expectation the closed loop already computes by rollout). Ours would be a second such closed form.

What it buys, in one mechanism:

- **The lateral comfort term.** With lateral variance on the ego's predicted path, a 0.5 m clearance puts real probability mass on the cyclist while a 1.5 m clearance does not, so the expected deficit grades smoothly with clearance even though every mean trajectory clears. This is precisely the DRF's Gaussian cross-section, arrived at from our own side, and it inherits the DRF's empirical support on overtaking.
- **The distance–time effect.** With prediction variance growing with horizon (and hence with distance at matched time), a far-but-time-matched conflict has a more dispersed predicted state. Under an asymmetric cost — which the preference function already has, since collisions cost far more than near-misses — expected deficit is *higher* for the more dispersed estimate. At matched PET the 50 km/h case is the nearer, and the 70 km/h case the more distant one; the sign of the resulting effect is exactly what needs checking against our data, and is the test proposed in §6.
- **A principled place for the epistemic half of the gaze machinery.** Uncertainty becomes a first-class quantity in the field rather than an unexercised branch.

Two honest caveats. First, adding predictive variance introduces at least one new parameter (how Σ scales), which erodes the "nothing upstream of the boundary is fitted" property — the same cost as the k decision in §5 of the overtake note, but larger. It should therefore be calibrated once, on one scenario, and frozen for the transfer test. Second, the *direction* of the distance effect under an expected-deficit model is a derivation I have sketched and not done; if the asymmetry runs the other way, the mechanism predicts the opposite of what we observe and the proposal fails. That derivation is the first task in §6, and it should be done before any code is written.

6 What should be checked, in order

- 1 **Derive the sign.** For our preference function's actual cost asymmetry, does $E[d(x)]$ under a wider Σ rise or fall? Analytic where possible, numeric otherwise. If the sign is wrong, stop; the mechanism is not the explanation.

[Done 2026-08-29 at review gate R.2, numerically on the actual preference function

(replication/czb/expected_deficit_sign.py → replication/czb/out/expected_deficit_sign.md), over the second cut-in study's own delta-velocity range. The outcome is worse for the proposal than the correction above feared, and different in character: the point field's within-row direction at matched TTC is inconsistent — $\rho(\text{gap}, d) = +1.00$ at TTC 2–3 s and -1.00 at TTC 5–7 s, through the counterfactual-residual term — where the data are uniformly negative (24 of 24 rows); and the distance-scaled noise moves $E[d]$ by at most ~10% of the mean deficit with inconsistent sign across rows. So the expected-deficit mechanism is a second-order, sign-unstable modulation where the data show a first-order monotone effect: it fails as the explanation of the distance–time anomaly, per this step's own stop rule. The lateral direction remains right, but see the gate record (docs/r2_gate_decisions.md) for why the lateral extension is not being built either: the field's longitudinal core was ruled against on the second cut-in study by the pre-registered field-versus-gap comparison, and steps 2 and 3 below are therefore not run — testing refinements of a core the data have rejected would be building on sand.]

- 1 **Test on LTAP, which is built for it.** The two-speed design separates time from distance by construction. Predicted: at matched PET, the expected-deficit field shows a speed effect of the observed sign and roughly the observed size, while the point-estimate field shows almost none.
- 2 **Test on the cyclist overtake.** Predicted: the expected-deficit field orders the 15 cells materially better than the current +0.402, approaching the clearance label's 0.833 without being given the clearance.
- 3 **Only then** consider whether the DRF's specific functional form (parabolic height, width linear in arc length and steering) should be adopted wholesale rather than re-derived, and whether its six constants can be inherited rather than refitted.

7 References

Bontje, F., van Waveren, F., van Maanen, L., Nallapu, B., Markkula, G., & Zgonnikov, A. (2026). *Knowing when to move: Evidence accumulation models of human behavior in traffic* (arXiv:2606.00727). arXiv.

<https://arxiv.org/abs/2606.00727>

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Mohammad, S. H. A., Farah, H., & Zgonnikov, A. (2024). *A cognitive process approach to modeling gap acceptance in overtaking* (arXiv:2306.05203). arXiv. <https://arxiv.org/abs/2306.05203>

Wang, Y., Srinivasan, A. R., Jokinen, J. P. P., Oulasvirta, A., & Markkula, G. (2024). *Pedestrian crossing decisions can be explained by bounded optimal decision-making under noisy visual perception* (arXiv:2402.04370). arXiv. <https://arxiv.org/abs/2402.04370>

Zgonnikov, A., Abbink, D., & Markkula, G. (2024). Should I stay or should I go? Cognitive modeling of left-turn gap acceptance decisions in human drivers. *Human Factors*, 66(5), 1399–1413. <https://doi.org/10.1177/00187208221144561>

Verification status, per the project's provenance rule. Kolekar et al. (2020), Bontje et al. (2026) and Mohammad et al. were read as full text (PDFs extracted locally); the DRF equations, parameter list and validated-scenario list in §3 are taken from the paper itself. Zgonnikov et al. (2024) and Wang et al. (2024) were read at publisher-page and abstract-plus-extract level respectively — their headline findings as quoted here are reliable, but the detailed noise-scaling argument attributed to Wang et al. in §2 has **not** been checked against their equations, and the year and venue of Mohammad et al. (arXiv preprint, also an IEEE conference paper) should be confirmed before any manuscript cites it.